$\mathbb{K}$ definitions as Matching Logic theories, formally

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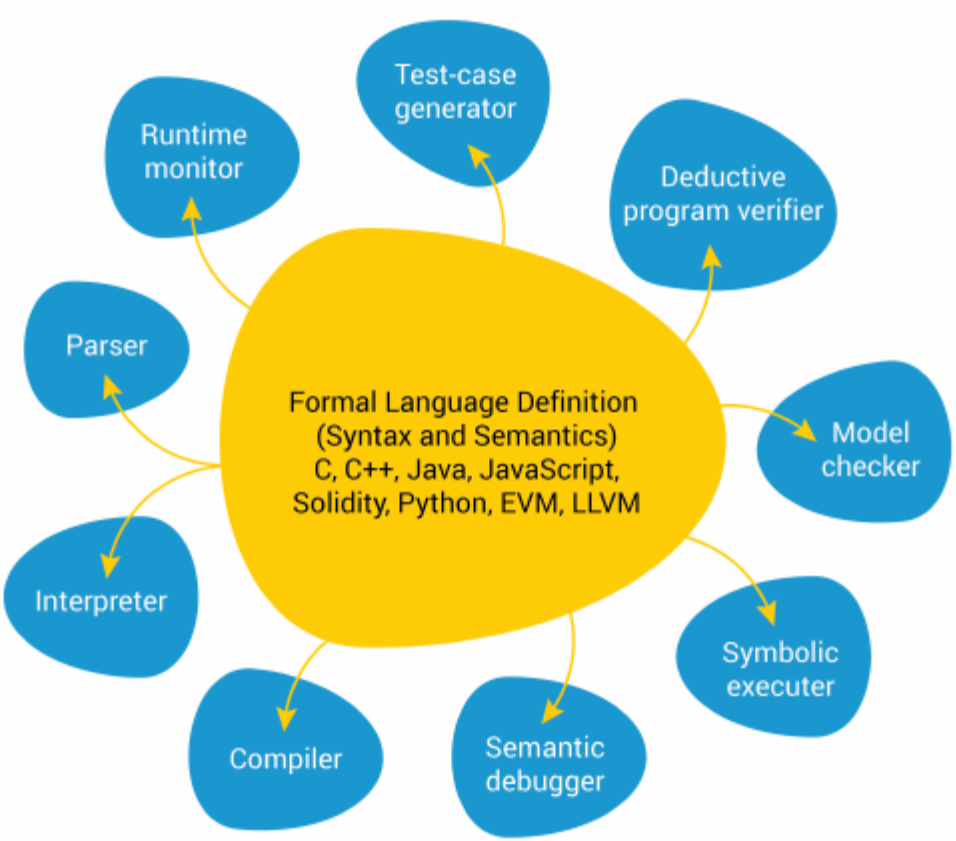
April 15, 2026,
FoSSaCS, Torino

𝕂 Framework

- Programming languages can be defined using a formal semantics based on rewriting
- Tools can be derived in a generic fashion from the PL definition



𝕂 Framework

```
module IMP-SYNTAX
  syntax AExp  ::= Int | Id
                 | "-" Int
                 > AExp "+" AExp    [left, seqstrict]
                 ...
  syntax Stmt  ::= Block
                 | Id "=" AExp ";"  [strict(2)]
                 ...
endmodule
module IMP
  imports IMP-SYNTAX
    configuration <T>
                    <k color="green"> $PGM:Pgm </k>
                    <state color="red">
                      .Map
                    </state>
                  </T>

  rule
    <state>... X |-> I ...</state>
    <k> X:Id => I ...</k>
  rule I1 + I2 => I1 +Int I2
  ...
endmodule
```

𝕂 Framework

```
int n, sum;
n = 0;
sum = 0;
n = sum + 1;
while ((n <= 100)) {
    sum = sum + n;
    n = n + 1;
}
```

```
> krun sum.imp --definition imp-kompiled --depth 114
<T>
  <k>
    sum = sum + n; ~> n = n + 1; ~> while (n <= 100) {
      sum = sum + n;
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    n |-> 4
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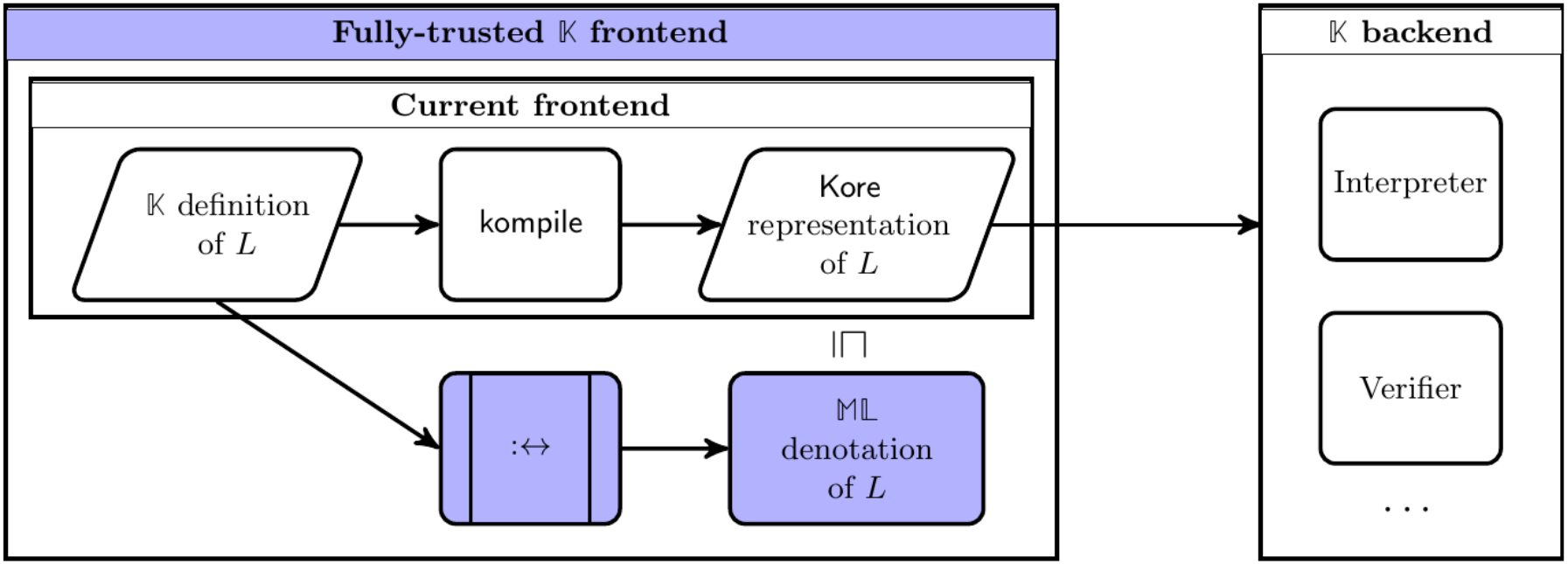
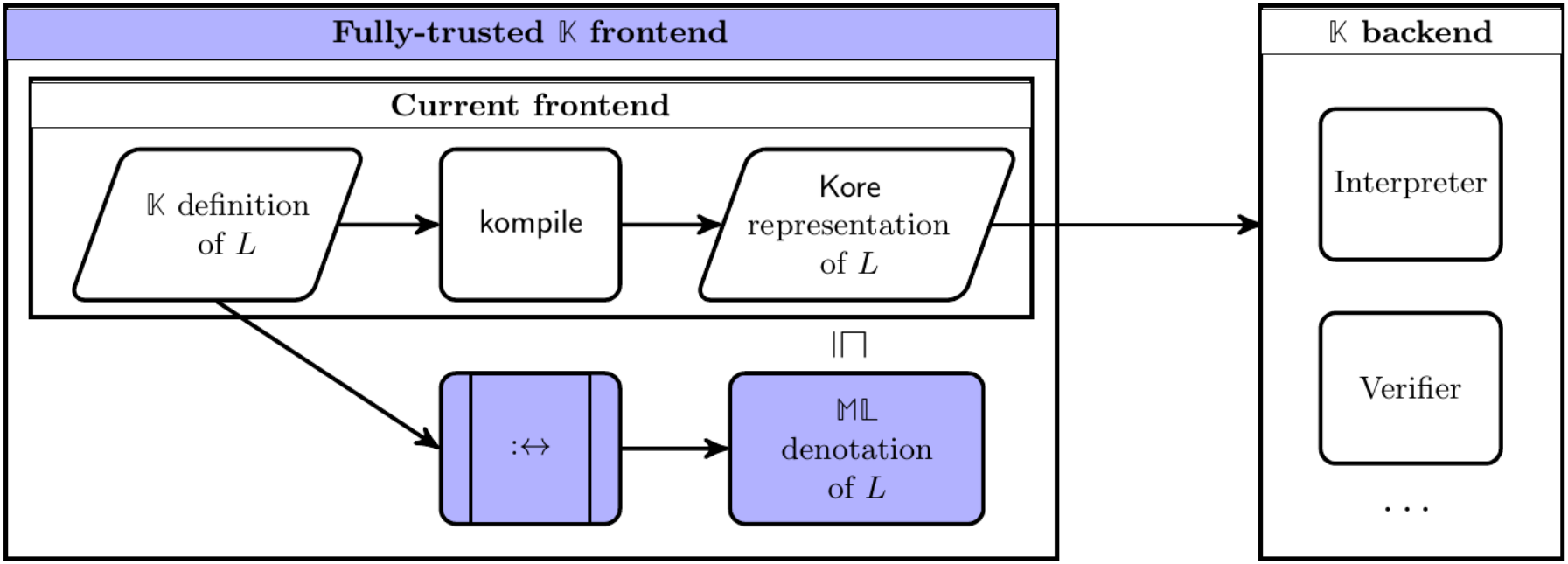
𝕂 Framework


Fully-trusted 𝕂 frontend
Current frontend
𝕂 definition of L
kompile
Kore representation of L
:↔
𝕄𝕃 denotation of L
𝕂 backend
Interpreter
Verifier
...

𝕂 Framework



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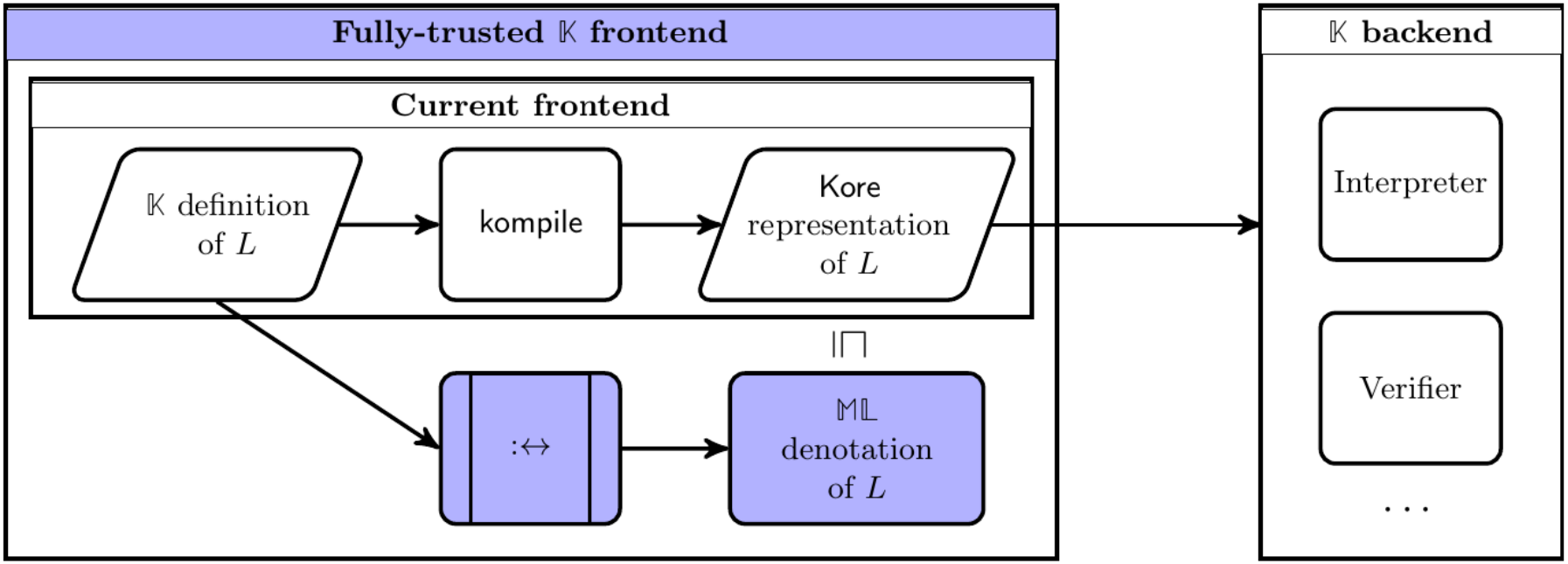

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Our contribution:

- A direct denotational semantics of 𝕂 in **Matching Logic**
- A local, composable axiomatization of **abstract rewrite rules**

Matching Logic

- Logical foundation of $\mathbb{K}$; minimal PL-focused system
- Versions: Many sorted ML (Chen, Roşu LMCS '17), Matching μ-Logic (Chen, Roşu LICS '19), Applicative ML (Chen, Roşu '19; Chen, Lucanu, Roşu JLAMP '21)

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Patterns, parametrized by a set of symbols Σ:

$$\varphi ::= x \mid X \mid \sigma \in \Sigma \mid \varphi\psi \mid \bot \mid \varphi \to \psi \mid \exists x.\varphi \mid \mu X.\varphi \quad \varphi \text{ positive for } X$$

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- $|\varphi|_\rho \subseteq M$
- Symbol σ denotes $M_\sigma \subseteq M$
- Application $\varphi\psi$ interpreted via $_\cdot_ : M \times M \to 2^M$
- Logical connectives denote set operations; μ is lfp operator

Matching Logic

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- **Theory of equality**:
  - Definedness: $\lceil _ \rceil$ with $\forall x. \lceil x \rceil$
  - Derived: $\lfloor \varphi \rfloor \equiv \neg \lceil \neg \varphi \rceil$, $\varphi = \psi \equiv \lfloor \varphi \leftrightarrow \psi \rfloor$, $\varphi \subseteq \psi \equiv \lfloor \varphi \rightarrow \psi \rfloor$, $x \in \varphi \equiv x \subseteq \varphi$

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  - Binders $\lambda x.\varphi$ encodable via power sorts (Chen, Roşu ICFP '20)

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- **Idea:** reify contexts as patterns of the form $\gamma\Box{:}s.\varphi$:

$$\begin{aligned} \gamma\Box{:}s.\varphi &:\equiv \mathsf{gamma}(\mathsf{intension}(\exists\Box.\,\langle\Box,\varphi\rangle)), \\ C[x] &:\equiv \mathsf{plug}(C,x) \end{aligned}$$

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- β-reduction for plugging can be derived:

$$(\gamma\Box{:}s.\varphi)[\psi] = \varphi[\psi/\Box]$$

- Composition $C_2 \circ C_1$: function composition on contexts

Configurations

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- k-cell: associative list of computations

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$$
\begin{aligned}
&\langle\!|\mathsf{T}|\!\rangle(\\
&\quad \langle\!|\mathsf{k}|\!\rangle(\mathsf{assign}(\mathit{sum}, \mathsf{plus}(\mathit{sum}, n)) \curvearrowright \mathsf{assign}(n, \mathsf{plus}(n, 1)) \curvearrowright w \curvearrowright \mathsf{dotK}),\\
&\quad \langle\!|\mathsf{state}|\!\rangle(n \mapsto 4 \ \mathsf{mapConcat}\ \mathit{sum} \mapsto 6)\\
&)
\end{aligned}
$$

Rule axiomatization: kompile

`kompile` output (roughly) for `IMP`:

$$\langle\!|\mathsf{T}|\!\rangle(\langle\!|\mathsf{k}|\!\rangle(\mathsf{plus}(i_1, i_2) \curvearrowright \kappa), s{:}\mathsf{Cell}\langle\mathsf{state}\rangle)$$

$$\Rightarrow$$

$$\langle\!|\mathsf{T}|\!\rangle(\langle\!|\mathsf{k}|\!\rangle(i_1 +_{\mathsf{Int}} i_2 \curvearrowright \kappa), s{:}\mathsf{Cell}\langle\mathsf{state}\rangle)$$

Rewriting: symbol $\bullet$ (**next**); $\varphi \Rightarrow \psi :\equiv \varphi \to \bullet\psi$.

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```
configuration
<T color="yellow">
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$\mathbb{ML}$ axiom:

$$\forall C\colon \mathsf{Context}^{\mathsf{Cell}\langle\mathsf{T}\rangle}_{\mathsf{Cell}\langle\mathsf{k}\rangle}.\forall i_1, i_2\colon \mathsf{Int}.\forall \kappa\colon \mathsf{K}.$$
$$(C \circ C_\kappa)[\mathsf{plus}(i_1, i_2)] \Rightarrow (C \circ C_\kappa)[i_1 +_{\mathsf{Int}} i_2],$$

where $C_\kappa :\equiv \gamma\Box\colon \mathsf{K}.\langle\!|\mathsf{k}|\!\rangle(\Box \curvearrowright \kappa)$.

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Importantly, the same axiom can be used in any configuration.

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$\mathbb{ML}$ axioms:

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$$(C \circ C_\kappa)[\mathsf{assign}(i, e)] \wedge \neg\,\mathsf{KResult}(e) \Rightarrow (C \circ C_\kappa)[e \curvearrowright C_{\mathsf{assign},2}]$$
$$\forall C\colon \mathsf{Context}^{\mathsf{Cell}\langle\mathsf{T}\rangle}_{\mathsf{Cell}\langle\mathsf{k}\rangle}.\forall \kappa\colon \mathsf{K}.\forall i\colon \mathsf{Id}.\forall e\colon \mathsf{AExp}.$$
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where

$C_\kappa \equiv \gamma\Box\colon \mathsf{K}.\langle\!|\mathsf{k}|\!\rangle(\Box \curvearrowright \kappa)$

$C_{\mathsf{assign},2} \equiv \gamma\Box\colon \mathsf{AExp}.\,\mathsf{assign}(i, \Box)$

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